

GIRINATH M

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CAREER OBJECTIVE

Electrical Engineering undergraduate bringing a unique blend of hardware-level systems thinking and software development skills in Java, SQL, and Python. Adaptable and quick to learn, seeking an entry-level software development role where embedded systems knowledge and programming ability intersect.

EDUCATIONAL QUALIFICATIONS

B.E., Electrical & Electronics Engineering, *V.S.B Engineering College, Karur, Tamil Nadu* | CGPA: 8.06/10 2023 – 2027
HSC, *M.S.P. Solai Nadar Memorial Boys HSS, Dindigul* | 74.83% 2021 – 2023

TECHNICAL SKILLS

Programming & Tools: Java, SQL, MATLAB/Simulink, VS Code, GitHub, Embedded C++
Platforms: Arduino IDE, Blynk IoT Cloud, SAP Build (Joule Studio)

PROJECTS

Smart Ambient Humidifier

Technologies: ESP32 Microcontroller (Embedded C++), DHT22 Humidity Sensor

- Developed a humidity control system that monitors real-time ambient humidity levels and automatically regulates moisture using sensor feedback.

IoT-Automated Water Treatment Reactor

Technologies: C++ (Arduino), ESP8266 (NodeMCU), Blynk IoT Cloud, Serial Communication

- Built a cloud-connected water treatment simulation with automated treating, settling, and skimming phases; synced physical hardware with a real-time web dashboard and automated overcurrent safety interlocks.

PAPER PUBLICATION

IoT-Based Grass Cutting Robot, *IEEE Xplore* 2026

- Built a solar-powered, IoT-based autonomous grass-cutting robot; published at the International Conference on Innovative Computing ([ICSES](#)).

INTERNSHIPS

Industrial Intern, *Tamil Nadu Newsprint and Papers Limited (TNPL)* 2025 | Pugalur

- Gained hands-on exposure to industrial automation, renewable energy conversion, power distribution, and heavy electrical machinery.

Industrial Intern, *Poppys Knit Wear Pvt. Ltd.* 2025 | Kangeyam

- Explored industrial textile manufacturing processes and the integration of sustainable solar energy systems for industrial power.

WORKSHOPS

Potential Applications of AI & ML in Electrical Engineering, *The Institution of Engineers (India), Tiruchirappalli* 07/2025
Recent Advances in Hybrid Energy Storage Management for EV Applications, *CSIR India* 09/2024

CERTIFICATES

- Python Foundation Certification — Infosys Springboard
- Introduction to IoT and Digital Transformation — Cisco & NASSCOM
- Exploring SAP S/4HANA Sales Essentials — SAP Learning Hub
- Introducing Joule Studio in SAP Build — SAP Learning Hub
- Micro-Certification - Agentic AI Executive— Servicenow

ACHIEVEMENTS

- Secured Cash Prize in Paper Presentation, "SENSORICS 2024" — Kongu Engineering College, Perundurai, Erode (2024)
- Capgemini Brand Quest — Campus Ambassador, Capgemini (2024)
- NSS Member — Organised and led initiatives in college club activities

LANGUAGES KNOWN

Tamil, English

INTERESTS

Electric Vehicle (EV) Technology, Software Development, IoT Application Development, Embedded Systems